V2 ChipSat Programming Guide

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1 Required Hardware/Software

V2 ChipSats use the STM32WL and are functionally just STM32 boards. The following lists the required hardware and software.

Hardware:

1. DeSCENT V2.X.X ChipSat
2. Pogo Programmer
3. ST-Link V2
4. Binder Clip (Optional)

Software

1. Arduino IDE
2. STM32CubeProgrammer
3. STM32 Arduino Core Documentation

2 Software Installation Steps

1. **Install STM32CubeProgrammer**
2. **Install STM32 Support in Arduino IDE**
   (a) Open **Arduino IDE** and go to `File → Preferences`.
   (b) In **Additional Board Manager URLs**, add:
   `https://github.com/stm32duino/BoardManagerFiles/raw/main/package_stmicroelectronics_index.json`
   (c) Open `Tools → Board → Boards Manager.`
   (d) Search for **STM32 MCU based boards** and install it.
3. **Wire ST-LINK to the Pogo Programmer**

   The Pogo Programmer has five pins on the right hand side populated with vertical header pins. Each are named accordingly.

ST-LINK Pin	Pogo Programmer Pin
SWDIO	SWDIO
SWCLK	SWCLK
GND	GND
3.3V	3.3V
NRST	NRST

There are some **special conditions** to connecting the **NRST Pin** that depend on the color of your ST-Link programmer. SSDS stocks two versions of the ST-Link Programmer, a Black and a Green Version. For the **Black Version** the Pogo Programmer NRST pin must be jumped to the ST-Link V2's RST pin for generally use (Post RDP reset). The **Green Version** can have the Pogo Programmer NRST pin left floating (Post RDP reset).

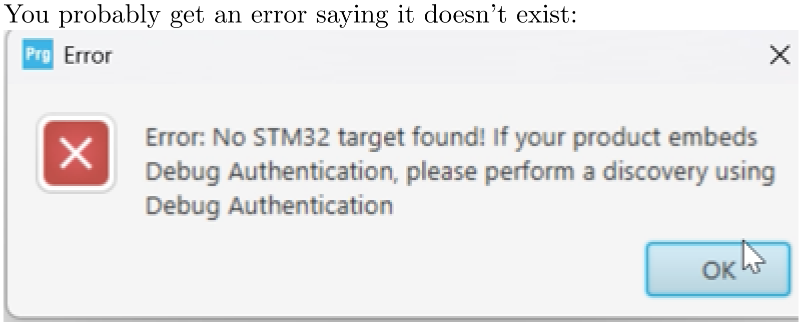
3 Configure Arduino IDE for STM32

- **Board:** LoRa Boards
- **Part Number:** LoRa-E5 mini
- **Upload Method:** STM32CubeProgrammer (SWD)

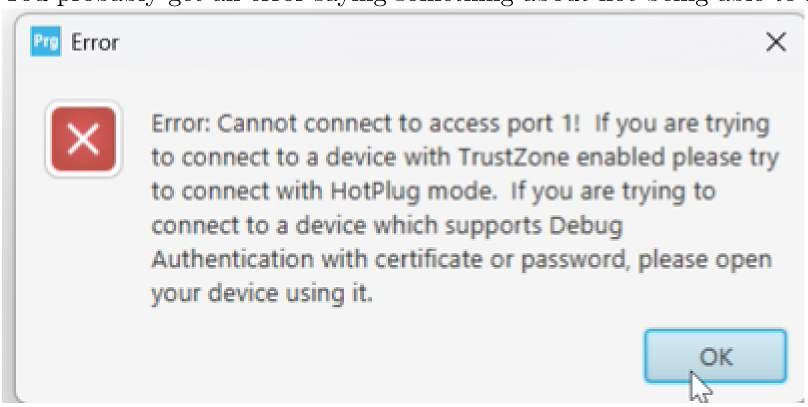
Generally, you can now upload sketches to the Wio now and skip directly to step 5. However, if your Wio is a newly shipped version and has never been programmed before, you must do the next step to ensure the data protection on the Wio is turned off before you are able to program it.

4 Connecting via STM32CubeProgrammer

1. Boot up STM32CubeViewer
2. Plug ST-Link in
3. Click connect on STM32CubeViewer
4. You probably get an error saying it doesn't exist:

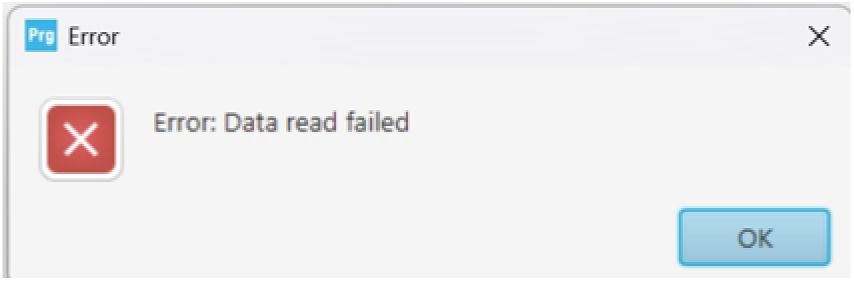


5. Take the NRST pin on the chipsat and short it to ground
6. Press Connect Again
7. You probably get an error saying something about not being able to read something:



8. Unplug NRST and leave it open
9. Press Connect Again

10. You probably get an error saying “Error: data read failed”



11. Go to the LHS and press OB (option bytes)
12. Change RDP to AA from BB
13. Press Apply
14. Now it works.

5 Uploading a test sketch

1. Open a basic example such as **Blink.**
2. Click **Upload.**
3. The sketch will compile and be uploaded via ST-LINK.

6 Codebase

Once you can get a blink sketch working, you can check our Github which includes functionality test’s for the chipsat: SSDS V1-V2